



RAMESHWAR SINGH THAKUR
CHAIRMAN, TAL
MANUFACTURING SOLUTIONS

Robotics

“ROBOTS Made In India COST LOWER Than Their Imported Counterparts”

TAL Manufacturing Solutions, a TATA Enterprise and a wholly owned subsidiary of TATA Motors, is currently the only manufacturer of commercial robots in India. It is also a notable producer of aerospace components. In a conversation with Paromik Chakraborty from Electronics For You, TAL chairman Rameshwar Singh Thakur talks about how robotics can benefit the manufacturing sector, the misconception about automation, and TAL's robot BRABO

Q. What are the benefits of using robotics in manufacturing?

A. The biggest intention of automation is to use robots in the dull, dirty and dangerous areas of work alongside multiplying the productivity and quality of yield. It is natural for human concentration and effort to fluctuate during strenuous, repetitive and risky work. Moreover, for human labour, you have sickness, marriage and other kinds of leaves of absence that might affect your delivery deadline, quantity of yield and quality of product.

It is important to understand that robots are a better proposition to take up tasks that require continuous precision and consistency, especially in hazardous environments. The aim is to assign robots to routine jobs while humans take up the high-level work. Till now, we have 60 robots in the industry and all of our customers have reported at least a 25 per cent jump in productivity coupled with improved quality.

Q. Could you please explain with some example?

A. There is an auto components manufacturer in Satara, Pune, which supplies parts to motorcycle manufacturers. Its plant had three foreign-made robots. We replaced all three

of them with two of our robots. The manufacturer achieved a 23 per cent increase in productivity after the replacement.

Q. Isn't robotics an expensive proposition?

A. Robots made in India cost lower than their imported counterparts. For instance, our robots are 40 per cent cheaper than those procured from overseas, as we manufacture every component, except the motor, domestically. Thus, our products can be serviced quickly, too. Today, our robots cost between ₹ 400,000 and ₹ 700,000 and target the SME sector.

Q. What's the impact of robotics adoption on employment worldwide?

A. Let's take the example of China. China has a large population but they do not have the problem of unemployment. They use 49 robots for every 10,000 workers. Compared to that, India has a record of only two robots for every 10,000 workers. Despite having more robots and a large population, China has no problem with jobs.

We can also look at countries like Korea, which uses 600 robots for every 10,000 workers. Korea is

the most competitive country in the world. Hyundai came into the industry much later than other automobile companies and today, it is rubbing shoulders with all of them. The Europeans too have more than 300 robots per 10,000 workers. People may say that Korea and Europe have less population and shortage of labour. But China does not have shortage of labour, and still it is going tremendously strong with the use of robots.

There is a mindset that robots will take away jobs. The truth is that if you are not competitive, you won't be able to get a job anyhow. Robots are making people competitive.

Q. Coming to your recently launched robot BRABO, what all it can do?

A. We have provided functions of dispensing, pick-and-place, machine tending, press tending, optical inspection and so on in BRABO. We are also able to integrate the product with anything. The things BRABO can't do right now are painting and welding. While we have no plans of any upgrade in BRABO as far as painting is concerned, it will be continually upgraded in other areas. The upgrades will all be customer driven. **EFY**